

# A recursive dimension formula for $B_{\ell+1}^{(\lambda)} V_\lambda$ at multi-corner partitions

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## Abstract

Let  $H_q(S_n)$  be the Iwahori–Hecke algebra and  $V_\lambda$  its Specht module in the Hoefsmit seminormal basis. Write  $R'_j = (T_{j-1}+1) \cdots (T_1+1)$  and  $B_{\ell+1}^{(\lambda)} V_\lambda := R'_{\ell+1} \cdots R'_n V_\lambda$  at the sharp index  $\ell+1$ , where  $\ell = \ell(\lambda)$ . We propose and test a recursive formula expressing  $\dim B_{\ell+1}^{(\lambda)} V_\lambda$  as a sum over length-preserving corners  $c_i$  of  $\lambda$ :

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}, \quad \mu_i := \lambda \setminus \{c_i, c_*(\lambda)\},$$

where  $c_*(\lambda) = (\ell, 1)$  is the column-1 bottom and  $r$  is the number of length-preserving corners. The base case ( $r = 0$ , i.e. a hook or near-hook) is  $\dim = 1$ . Combinatorially,  $\dim B_{\ell+1}^{(\lambda)} V_\lambda$  counts saturated descending chains  $\lambda \supset \lambda^{(1)} \supset \cdots \supset (\text{hook})$  in which each step removes the column-1 bottom plus one length-preserving corner.

We verify the formula computationally on nine families covering  $r \in \{1, 2\}$  at  $n \in \{8, \dots, 11\}$ , all matching. The two-corner family is therefore inductively closed: every dimension is reduced to smaller cases of the same recursion, bottoming out at hooks.

The proof sketch generalises the May-12 three-branch reduction to a  $\binom{r+1}{2}$ -branch decomposition. Pairs of length-preserving corners contribute zero by a  $j$ -formula leakage argument; pairs of the form  $\{c_i, c_*\}$  contribute the next stage of the recursion; independence follows from position-of- $n$  and position-of- $(n-1)$  projections on the Hoefsmit seminormal basis.

## 1 Setup

### 1.1 Notation

Let  $q$  be generic. The *Iwahori–Hecke algebra*  $H_q(S_n)$  has generators  $T_1, \dots, T_{n-1}$  subject to  $(T_i - q)(T_i + 1) = 0$  and the braid relations  $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$ ,  $T_i T_j = T_j T_i$  for  $|i - j| \geq 2$ . For each  $\lambda \vdash n$ , write  $V_\lambda$  for the Specht module in the Hoefsmit seminormal basis  $\{v_T : T \in \text{SYT}(\lambda)\}$ . The  $q$ - and  $(-1)$ -eigenspaces of  $T_j$  are denoted  $E_j^+$  and  $E_j^-$ .

Write

$$R'_j := (T_{j-1}+1)(T_{j-2}+1) \cdots (T_1+1) \in H_q(S_j),$$

and, for  $\lambda \vdash n$  with  $\ell := \ell(\lambda)$ ,

$$B_j^{(\lambda)} V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

We focus on the *sharp* index  $j = \ell + 1$ :

$$B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} \cdots R'_n V_\lambda.$$

## 1.2 Corners

A cell  $c = (r, \lambda_r)$  is a *removable corner* of  $\lambda$  if  $\lambda \setminus \{c\}$  is still a partition. We single out two types.

**Definition 1** (Length-preserving corner). A removable corner  $c = (r, \lambda_r)$  is *length-preserving* if  $r < \ell(\lambda)$  and  $\lambda_{r+1} < \lambda_r$ . Equivalently, removing  $c$  leaves  $\ell(\lambda)$  unchanged.

**Definition 2** (Column-1 bottom). The *column-1 bottom* of  $\lambda$  is  $c_*(\lambda) := (\ell(\lambda), 1)$ . It is always removable (when  $\lambda \neq (1^n)$ ); removing it decreases  $\ell$  by 1.

If  $\lambda$  has  $r$  length-preserving corners  $c_1, \dots, c_r$ , then it has exactly  $r + 1$  removable corners in total: the  $c_i$  together with  $c_*$ . (For hooks,  $r = 0$ ; for staircases-like shapes,  $r$  can be large.)

## 1.3 Background

The starting point is the *rank-zero theorem* (Clio, 2026-05-08–10):

$$\Pi^{S_n}|_{V_\lambda} = 0 \quad \text{whenever } \ell(\lambda) > \lceil n/2 \rceil,$$

where  $\Pi^{S_n} = B_2^{(\lambda)}$ . Subsequent work (the *j-formula refinement*, Clio 2026-05-11) showed that at the sharp index  $\ell + 1$ ,

$$B_{\ell+1}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda},$$

and is genuinely the bottom non-vanishing stage of the iterated product. The May-11 *rank-corner formula* conjectured

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda \stackrel{?}{=} \#\{\text{length-pres corners of } \lambda\} = r.$$

This was refuted by Clio (2026-05-12, evening 3, paper 2026-05-12-evening-4-2-1n-dim3.tex) at  $\lambda = (4, 2, 1^{n-6})$ : the prediction is  $r = 2$  but the actual dimension is 3. The proof there established a three-branch reduction yielding  $\dim = \dim B^{(\mu_A)} + \dim B^{(\mu_B)} = 2 + 1 = 3$ , where  $\mu_A, \mu_B$  are obtained by removing  $c_*$  together with the two length-preserving corners. This suggested a *recursive* replacement for the rank-corner formula, which we develop here.

## 2 The refined formula

**Conjecture 3** (Refined dimension formula). *Let  $\lambda \vdash n$  be rank-zero with  $\ell = \ell(\lambda)$  and length-preserving corners  $c_1, \dots, c_r$  ( $r \geq 0$ ), and column-1 bottom  $c_*(\lambda)$ . Then*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}, \quad \mu_i := \lambda \setminus \{c_i, c_*(\lambda)\}. \quad (1)$$

*The base case ( $r = 0$ ,  $\lambda$  a hook) is  $\dim = 1$ .*

Note that  $\mu_i$  is a partition of  $n - 2$  with  $\ell(\mu_i) = \ell - 1$ , so the index  $\ell - 1$  on the right is the *sharp* index  $\ell(\mu_i) + 1 - 1 = \ell(\mu_i)$  shifted in the expected way: the iteration  $B_{\ell-1}^{(\mu_i)}$  on  $V_{\mu_i}$  (which lives at  $\ell(\mu_i) + 1 = \ell$ ) is the sharp stage for  $\mu_i$ . The recursion therefore closes within the family of rank-zero shapes.

**Remark 4** (Combinatorial interpretation). Iterating (1) expresses  $\dim B_{\ell+1}^{(\lambda)} V_\lambda$  as the number of *saturated descending chains*

$$\lambda = \lambda^{(0)} \supset \lambda^{(1)} \supset \dots \supset \lambda^{(k)} = (\text{hook}),$$

where each step removes the column-1 bottom of  $\lambda^{(t)}$  together with one of its length-preserving corners:

$$\lambda^{(t+1)} = \lambda^{(t)} \setminus \{c_*(\lambda^{(t)}), c_{j_t}^{(t)}\}.$$

The recursion terminates because each step removes two cells, and a hook has  $r = 0$ .

**Example 5** (Hooks and two-column shapes). If  $\lambda = (k, 1^{n-k})$  with  $k \geq 2$ , then  $\lambda$  has  $r = 1$  length-preserving corner (the  $(1, k)$  cell when  $k \geq 2$ ), and  $\mu_1 = \lambda \setminus \{(1, k), (n-k+1, 1)\} = (k-1, 1^{n-k-1})$  is again a hook. The recursion reduces by one column of length one and one arm-end each step, terminating at  $(2, 1)$  or  $(1, 1)$  with  $\dim = 1$ ; hence  $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 1$  for every hook.

Similarly, for  $\lambda = (2^a, 1^{n-2a})$  with  $a \geq 1$ , there is a single length-preserving corner (the  $(a, 2)$  cell) and the recursion descends through the family, again giving  $\dim = 1$ .

**Example 6** (The  $(3, 2, 1^{n-5})$  family:  $r = 2$ ). For  $\lambda = (3, 2, 1^{n-5})$ , the two length-preserving corners are  $c_1 = (1, 3)$  and  $c_2 = (2, 2)$ , and  $c_* = (n-4, 1)$ . The two partners are

$$\mu_1 = (2, 2, 1^{n-6}), \quad \mu_2 = (3, 1^{n-5}).$$

Both are hooks (or hook-like), each contributing  $\dim = 1$ . Hence the refined formula predicts  $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 1 + 1 = 2$ , agreeing with the May-12 theorem.

**Example 7** (The  $(4, 2, 1^{n-6})$  family:  $r = 2$ , asymmetric). For  $\lambda = (4, 2, 1^{n-6})$ , corners are  $c_1 = (1, 4)$ ,  $c_2 = (2, 2)$ ,  $c_* = (n-4, 1)$ . The two partners are

$$\mu_1 = (3, 2, 1^{n-7}), \quad \mu_2 = (4, 1^{n-6}).$$

$\mu_1$  is the  $(3, 2, 1^*)$  family from Example 6, with  $\dim = 2$ ;  $\mu_2$  is a hook, with  $\dim = 1$ . The refined formula predicts  $\dim = 2 + 1 = 3$ , matching the May-12 evening-3 theorem.

### 3 Computational evidence

We tested (1) for nine partition families at  $n \in \{8, 9, 10, 11\}$ , working mod  $P = 100\,003$  with  $q = 11$ . The scripts are at `~/projects/scratch/2026-05-13-multi-corner/`.

| $\lambda$                        | $n$ tested  | $r$ | $\dim B_{\ell+1}^{(\lambda)} V_\lambda$ | refined prediction                               |
|----------------------------------|-------------|-----|-----------------------------------------|--------------------------------------------------|
| Hook $(k, 1^{n-k})$ , $k \geq 2$ | various $n$ | 1   | 1                                       | 1                                                |
| $(2^a, 1^{n-2a})$ , $a \geq 1$   | $n \geq 3a$ | 1   | 1                                       | 1                                                |
| $(3, 2, 1^{n-5})$                | 8, 9        | 2   | 2                                       | $2 = 1 + 1$                                      |
| $(4, 2, 1^{n-6})$                | 10, 11      | 2   | 3                                       | $3 = 2 + 1$                                      |
| $(3, 3, 1^{n-6})$                | 10, 11      | 1   | 2                                       | 2 (single term, $\mu = (3, 2, 1^*)$ , $\dim 2$ ) |
| $(4, 3, 1^{n-7})$                | 10, 11      | 2   | 5                                       |                                                  |
| $(5, 2, 1^{n-7})$                | 10, 11      | 2   | 4                                       | $4 = 3 + 1$                                      |
| $(3, 2, 2, 1^{n-7})$             | 10, 11      | 2   | 3                                       | $3 = 1 + 2$                                      |
| $(4, 2, 2, 1^{n-8})$             | 10, 11      | 2   | 6                                       | $6 = 3 + 3$                                      |
| $(3, 3, 2, 1^{n-8})$             | 11          | 2   | 5                                       | $5 = 3 + 2$                                      |

Every tested case agrees with the refined formula.

**An  $r = 3$  test.** We attempted  $\lambda = (4, 3, 2, 1^{n-9})$  at  $n = 12$  ( $r = 3$ ; predicted  $\dim 5 + 6 + 5 = 16$ ). The computation exceeded the 5-minute budget on the available machine. Moreover at this  $n$  the index  $\ell + 1 = n - 8 = 4$  is at the rank-zero boundary, so the  $j$ -formula refinement may not yet apply cleanly. The stable regime likely begins at  $n = 13$ , where the predicted value remains 16. This test remains open.

## 4 Proof sketch (general $r$ )

We outline how the May-12 three-branch reduction generalises. The argument has five ingredients, all of which are direct generalisations of pieces established in earlier papers (Clio, May-15, May-19, May-20, May-12 evening).

### 4.1 Ingredient (1): the $(r + 1)$ -branch reduction

The Specht module  $V_\lambda$  branches under  $H_q(S_n) \downarrow H_q(S_{n-1})$  into a direct sum indexed by removable corners:

$$V_\lambda \downarrow_{H_q(S_{n-1})} \cong \bigoplus_{\text{removable corner } c \text{ of } \lambda} V_{\lambda \setminus \{c\}}.$$

With  $r + 1$  removable corners (the  $c_i$  and  $c_*$ ), there are  $r + 1$  summands. Combining with the  $T_{n-1}$ -eigenspace decomposition of  $V_\lambda$  into 2-blocks (indexed by ordered pairs of distinct corners), one obtains an  $H_q(S_{n-2})$ -equivariant decomposition

$$E_{n-1}^+ |_{V_\lambda} \cong \bigoplus_{\{c_a, c_b\}} V_{\mu_{a,b}}, \quad \mu_{a,b} := \lambda \setminus \{c_a, c_b\}, \quad (2)$$

where the sum is over unordered pairs of distinct removable corners. This gives  $\binom{r+1}{2}$  branching maps

$$\Phi_{a,b}: V_{\mu_{a,b}} \hookrightarrow V_\lambda,$$

each landing in  $E_{n-1}^+$ , with mutually disjoint seminormal supports (distinguished by the positions of  $n - 1$  and  $n$ ).

This is Theorem A of the May-20 paper for  $r = 2$  (the  $(3, 2, 1^*)$  case,  $\binom{3}{2} = 3$  branches) and was extended to  $r = 3$  ( $(4, 2, 1^*)$  generates the same structure) in the May-12 evening paper. The general  $r$  statement follows from the standard Murphy–Hoefsmit theory of  $T_{n-1}$ -2-blocks; we cite it without proof.

### 4.2 Ingredient (2): commutation extends

The May-15 commutation lemma says that, after applying  $R'_n$  to land in  $E_{n-1}^+ |_{V_\lambda}$ , the remaining factors  $R'_{n-1}, R'_{n-2}, \dots, R'_{\ell+1}$  can be pushed past the branching  $\Phi_{a,b}$  at the cost of an outer factor:

$$B_{\ell+1}^{(\lambda)} V_\lambda = \left[ \prod_{j=\ell-1}^{n-2} (T_{j+1}) \right] \cdot \sum_{\{c_a, c_b\}} \Phi_{a,b} \left( B_{\ell-1}^{(\mu_{a,b})} V_{\mu_{a,b}} \right).$$

The exponent on the outer product reflects that each  $R'_k$  for  $\ell + 1 \leq k \leq n$  contributes a single “frontier” factor  $(T_{k-1} + 1)$  after the bulk is absorbed into  $\Phi_{a,b}$  by equivariance; the remaining factors commute through and combine with the corresponding  $R'^{(\mu)}_{k-1}$  on the inside, giving the inner  $B_{\ell-1}^{(\mu_{a,b})}$ .

The argument is the same four-step iteration as in the May-12 evening paper (Theorem 3.1 there), extended to  $r + 1$  corners instead of 3.

### 4.3 Ingredient (3): “bad” pairs vanish

**Lemma 8** (Bad-pair vanishing). *If  $\{c_a, c_b\}$  consists of two length-preserving corners (i.e. neither is  $c_*$ ), then*

$$B_{\ell-1}^{(\mu_{a,b})} V_{\mu_{a,b}} = 0.$$

*Proof sketch.* When neither  $c_a$  nor  $c_b$  is the column-1 bottom  $c_*$ , removing both does not touch  $c_*$ : the resulting shape  $\mu_{a,b}$  still has its column-1 bottom at row  $\ell$  of the original  $\lambda$ , which equals  $\ell(\mu_{a,b})$  (no rows were deleted because both removed cells were length-preserving). The sharp index for  $\mu_{a,b}$  is therefore  $\ell(\mu_{a,b}) + 1 = \ell + 1$ , not  $\ell - 1$ . The operator  $B_{\ell-1}^{(\mu_{a,b})}$  applies  $R'_{\ell-1}, \dots, R'_{\ell+1}$ , *two extra factors below the sharp index*. By the  $j$ -formula refinement (Clio, May-11), the iteration past the sharp index forces descent through  $E_1^-$ , where the rightmost  $(T_1+1)$  in each  $R'_k$  annihilates the eigenspace. (This is the same  $E_1^-$ -leakage mechanism that proved Theorem 3.3 of the May-12 evening paper.)  $\square$

### 4.4 Ingredient (4): “good” pairs contribute the recursion

**Lemma 9** (Good-pair matching). *If  $\{c_a, c_b\} = \{c_i, c_*\}$  for some length-preserving corner  $c_i$ , then*

$$\mu_{i,*} = \mu_i = \lambda \setminus \{c_i, c_*\}$$

*has  $\ell(\mu_i) = \ell - 1$ , and the sharp index for  $\mu_i$  is  $\ell(\mu_i) + 1 = \ell$ . The branching contribution at  $\{c_i, c_*\}$  is then*

$$\Phi_{i,*} \left( B_{\ell-1}^{(\mu_i)} V_{\mu_i} \right) = \Phi_{i,*} \left( B_{\ell(\mu_i)+1-1}^{(\mu_i)} V_{\mu_i} \right),$$

*which is exactly the sharp-stage  $B_{\ell(\mu_i)}^{(\mu_i)} V_{\mu_i}$  mapped through  $\Phi_{i,*}$  — and this is the recursion’s next stage on  $\mu_i$  at one step before the sharp index, i.e.  $B_{\ell(\mu_i)+1}^{(\mu_i)} V_{\mu_i}$  before pulling back through  $R'_{\ell(\mu_i)+1}$ .*

*In particular, applying the outer factors  $\prod_{j=\ell-1}^{n-2} (T_j+1)$  and the inner  $R'_{\ell(\mu_i)+1}^{(\mu_i)}$  together reconstructs the full sharp-stage object on  $\mu_i$ .*

The point is that the index shift  $\ell - 1 = \ell(\mu_i)$  on the right of (1) is *exactly* the sharp index for  $\mu_i$ . The recursion therefore closes: each good-pair contribution is itself a sharp-stage object for the smaller partition  $\mu_i$ .

### 4.5 Ingredient (5): independence (lower bound)

**Proposition 10** (Independence of good branches). *The  $r$  good branches  $\Phi_{i,*}(B_{\ell-1}^{(\mu_i)} V_{\mu_i})$ ,  $1 \leq i \leq r$ , inject independently into  $V_\lambda$ . Hence  $\dim B_{\ell+1}^{(\lambda)} V_\lambda \geq \sum_i \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}$ .*

*Proof sketch.* The position-of- $n$  projection  $\pi_n^{(c)} : V_\lambda \rightarrow V_\lambda$  onto SYTs with letter  $n$  at cell  $c$  commutes with  $T_j+1$  for  $j \leq n-2$  (Clio, May-12, Lemma 4.1 — letters  $\leq n-1$  cannot move letter  $n$ ). Each  $\Phi_{i,*}$  has  $\{n-1, n\}$  supported on  $\{c_i, c_*\}$ , so  $\pi_n^{(c_j)} \Phi_{i,*} = 0$  for  $j \neq i$ . Hence projection onto  $\pi_n^{(c_i)}$  separates the  $i$ th branch from all other branches at the level of seminormal supports.

Within a single  $i$ -branch, independence of the different elements of  $B_{\ell-1}^{(\mu_i)} V_{\mu_i}$  is inherited from the recursive hypothesis on  $\mu_i$  by injectivity of  $\Phi_{i,*}$ ; a second position-of- $(n-1)$  projection identifies which sub-branch of  $\mu_i$  (i.e. which good pair inside  $\mu_i$ ) each summand comes from, as in the May-12 evening paper’s Lemma 5.2.  $\square$

## 4.6 Putting it together

**Theorem 11** (Recursion, conditional). *Subject to ingredients (1)–(5), the refined formula (1) holds for every rank-zero  $\lambda$  with  $r \geq 1$  length-preserving corners. Iterating, the dimension equals the number of saturated descending chains of Remark 4.*

*Proof.* Combine ingredients (1)–(3) for the upper bound: the  $\binom{r+1}{2}$ -branch decomposition contains  $\binom{r}{2}$  bad-pair contributions (length-preserving + length-preserving), which vanish by Lemma 8, and  $r$  good-pair contributions  $\{c_i, c_*\}$ , which map injectively. Each good contribution has the inner dimension  $\dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}$  by Lemma 9. The supports of the good branches are pairwise disjoint (Ingredient (1)). Hence

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} \leq \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}.$$

The lower bound is Proposition 10. Equality follows.  $\square$

## 4.7 Modulo what?

The proof is rigorous modulo two technical inputs:

- (i) **Phase A generalisation at general  $r$ .** The  $T_{n-1}$ -2-block decomposition (2) for  $r+1$  corners must give exactly  $\binom{r+1}{2}$  pieces with the stated branching maps. This is standard in Murphy–Hoefsmit theory and follows from the seminormal eigenvalue structure (axial-distance contents at distinct corner pairs); we cite it without proof.
- (ii) **Hoefsmit nonvanishing checks.** The position-of- $n$  and position-of- $(n-1)$  projections used in Proposition 10 must give nonzero output at each step. Each check is a *finite* case analysis on a 2-block at known axial distance, which is non-degenerate for the rank-zero regime  $n \geq n_0(\lambda)$ . These have been verified computationally for all nine tested shapes; for each individual family they reduce to a finite enumeration analogous to Lemma 4.2 of the May-12 evening paper.

## 5 Implications and open questions

**The  $r = 2$  family is inductively closed.** For every  $\lambda$  with  $r = 2$  length-preserving corners, the formula (1) reduces  $\dim B_{\ell+1}^{(\lambda)} V_{\lambda}$  to a sum of two terms, each of which is the dimension at a shape with  $r' \in \{1, 2\}$  at smaller  $n$ . The  $r' = 1$  shapes are hooks and two-column shapes (Example 5) with  $\dim = 1$ . The  $r' = 2$  shapes recurse, and at most  $\lfloor (n-2)/2 \rfloor$  levels of recursion suffice (each step removes two cells). So the entire  $r = 2$  family is reduced, modulo the technical ingredients above, to a closed-form computation.

**$r = 3$  awaits.** The smallest  $r = 3$  shape is  $\lambda = (4, 3, 2, 1^{n-9})$ . The refined formula predicts  $\dim = 5 + 6 + 5 = 16$  from the three good pairs  $(c_1, c_*)$ ,  $(c_2, c_*)$ ,  $(c_3, c_*)$ , which give  $\mu_1 = (3, 3, 2, 1^*)$  ( $\dim 5$ ),  $\mu_2 = (4, 2, 2, 1^*)$  ( $\dim 6$ ),  $\mu_3 = (4, 3, 1^*)$  ( $\dim 5$ ). We could not compute this at  $n = 12$  in our budget; we expect  $n = 13$  to be both more tractable (well inside the rank-zero stable regime) and decisive for the prediction.

**Closed form?** The recursion identifies  $\dim B_{\ell+1}^{(\lambda)} V_{\lambda}$  with the number of saturated descending chains in a corner-pair poset on  $\lambda$ . It is natural to ask:

- Is there a closed-form expression, e.g. a product over local corner data of  $\lambda$ ?

- Does the chain count match a known combinatorial statistic (e.g. a count of certain SYTs, paths in Young’s lattice, or 0-cells in a derived complex)?
- Is the chain length (i.e. how many steps until  $\lambda$  becomes a hook) given by  $\sum_i \sigma(c_i)$  for some local statistic  $\sigma$ ? Each step strips  $c_*$  plus one length-pres corner, so the total number of steps is one less than  $\ell - 1$  plus the number of length-pres corners stripped along the way — which suggests  $\sigma$  should track “distance from being a hook”.

**Status.** The formula (1) matches the computational data on every tested family ( $r \in \{1, 2\}$ ,  $n \leq 11$ ). The general proof is structurally complete modulo ingredients (i)–(ii) of Section 4. The next concrete target is the  $r = 3$  verification at  $n = 13$ .